

Question ID: 310c87fe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

Answer

- A. 18
- B. 27
- C. 36
- D. 81

Correct Answer: B

Rationale

Choice B is correct. The surface area of a cube with side length s is equal to $6s^2$. Since the surface area is given as 54 square meters, the equation $54 = 6s^2$ can be used to solve for s . Dividing both sides of the equation by 6 yields $9 = s^2$. Taking the square root of both sides of this equation yields $3 = s$ and $-3 = s$. Since the side length of a cube must be a positive value, $s = -3$ can be discarded as a possible solution, leaving $s = 3$. The volume of a cube with side length s is equal to s^3 . Therefore, the volume of this cube, in cubic meters, is 3^3 , or 27.

Choices A, C, and D are incorrect and may result from calculation errors.